PCDA: Pseudo-Label Correction for Source-Free Universal Domain Adaptation in Remote Sensing Scene Classification

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***Abstract*—The precise classification of remote sensing images is crucial for applications such as disaster monitoring, land use planning, and national security. Since effective classification models often rely on large, well-annotated datasets, the remote sensing field faces two persistent challenges: the limited availability of labeled data and the domain shifts that hinder model generalization. Existing approaches primarily include data generation-based domain adaptation (DA) schemes and large model-based zero-shot learning methods. The former synthesizes target domain samples via generative adversarial networks (GANs) to address the source-free DA (SF-DA) problem, but consequently introduces significant noise interference, and the latter excessively relies on the generalization capability of large models; this capability is constrained by the inherent long-tailed distribution characteristic in remote sensing scene classification tasks, failing to be fully exploited. To overcome these limitations, we propose pseudo-label correction for source-free universal DA (PCDA), a novel framework that addresses both data privacy and domain/category shifts more effectively without source data. We propose generating class-strategy-adaptive pseudo-prototypes instead of dataset generation to reduce training costs while introducing source domain knowledge. And design an implicit pseudo-label correction mechanism that dynamically optimizes pseudo-labels, effectively reducing the interference of pseudo-label uncertainty in model training, thereby enhancing model robustness. Experimental results demonstrate that the proposed method achieves remarkable average classification accuracies of 88.1%, 82.7%, 71.7%, and 75.6% in the R-U, R-A, R-N, and A-N dataset pairs, respectively. The code is available at https://github.com/TJU-L-Ning/PCDA**

***Index Terms*—Pseudo-label correction, remote sensing scene classification, source-free domain adaptation (SF-DA), universal domain adaptation (UniDA).**

NOMENCLATURE

D_s	Source domain.
D_t	Target domain.
L_t	Target label set, which is inaccessible in train stage.
Φ_s	Model trained on the source domain.
Φ_t	Model trained on the unlabeled target domain with proposed PCDA.
$\overline{C_s}$	Source classifier, whose parameter is frozen.
n^s	Counts of source categories.
n^t	True counts of target categories.
x^t	Unlabeled image from target domain.
$\vec{P}_{si}$	Positive prototype of the i_{th} source category.
$\vec{N}_{si}$	Negative prototype of the i_{th} source category.
h	Predicted number of target classes.
M_K^+	Positive sample set.
m_k^i	Confidence gap between the *top* 1 and *top* 2 predictions for the i_{th} class.
R_k	Pseudo-margin value for the k_{th} category.
γ	Bias introduced to control the effect brought by pseudo-margin.
τ^k	Size of positive sample set under the k_{th} class.
ξ	Jaccard index.
N_t	Number of samples in the target domain.
C_o^k	Binary classifier under the k_{th} category.
$\vec{f}$	Feature vector of the image.
q_k^j	Discrete binary pseudo-labels composed of "0" and "1".
Q	Similarity matrix.
$A(\cdot\ ,\cdot)$	Confidence function.
$D(\cdot\ ,\cdot)$	Cosine distance function.
D_{KL}	Kullback–Leibler divergence.
λ	Adjustment coefficient.

I. INTRODUCTION

REMOTE sensing scene classification plays a crucial role in various real-world applications, such as environmental monitoring, urban planning, disaster response, and agricultural management [1], [2], [3], [4]. The rapid development of deep learning has significantly advanced this field, especially when large-scale annotated datasets are available [5], [6]. While most academic benchmark models are released alongside public training datasets to support research reproducibility, full access to raw source data is rarely feasible in real-world remote sensing deployments, due to geospatial confidentiality regulations, cross-institutional data sharing constraints, and prohibitive storage and transmission costs of large-scale imagery.

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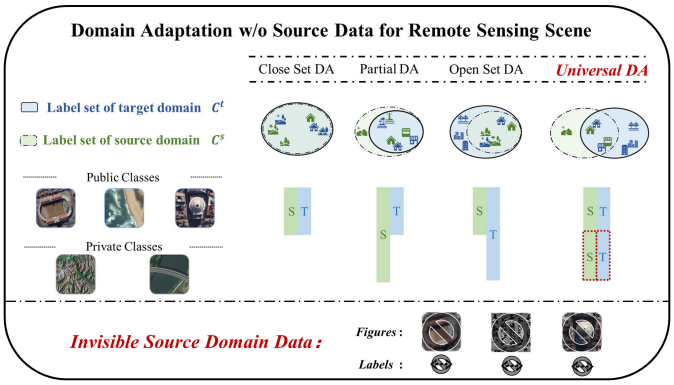


Fig. 1. Four types of SF-DA settings in remote sensing scene classification. Unlike setups relying on known label set relationships, SF-UniDA assumes unknown overlap between source and target label sets—posing the greatest challenge: identifying shared classes and rejecting private target classes without accessing source data.

To alleviate the reliance on labeled data, *unsupervised domain adaptation* (UDA) has emerged as a promising direction by transferring knowledge from a labeled source domain to an unlabeled target domain [7]. Yet, conventional UDA methods typically assume full access to the source data and a shared or partially known label space—assumptions that are often untenable in remote sensing practice. Data-sharing restrictions, security concerns, and unknown label mismatches between source and target domains motivate a more realistic and stringent setting: *source-free universal domain adaptation* (SF-UniDA). In this setting, models must adapt to an unlabeled target domain without access to source data and without prior knowledge of label space overlap. As illustrated in Fig. 1, compared with other UDA tasks that do not access source data, SF-UniDA imposes more relaxed requirements on the relationship between the source and target label sets, which makes it more naturally applicable to complex real-world scenarios.

Under the SF-UniDA setting, two key challenges arise. First, the absence of source data makes it difficult to align domain distributions. To address this, some prior works synthesize source-like samples using generative models [8], [9] or attempts of pseudo-label-based schemes in the field of remote sensing image segmentation [10]. While effective in approximating the source distribution, these approaches incur high computational costs and often suffer from training instability. Second, the lack of label space alignment introduces semantic uncertainty. Many existing methods rely solely on entropy-based thresholds to distinguish public and private classes [11] during the training stage.

The importance of robust generalization in this context is further emphasized by the emergence of large-scale vision-language models like CLIP [12]. Despite their success in natural image domains, we find that CLIP struggles with long-tailed label distributions in remote sensing scene datasets. As illustrated in Fig. 2, critical remote sensing categories tend to be underrepresented in CLIP's predictions, reflecting a gap between general-purpose pretraining and domain-specific semantics. This suggests that task-specific, lightweight models are still necessary to address the nuanced challenges of remote sensing domain shifts.

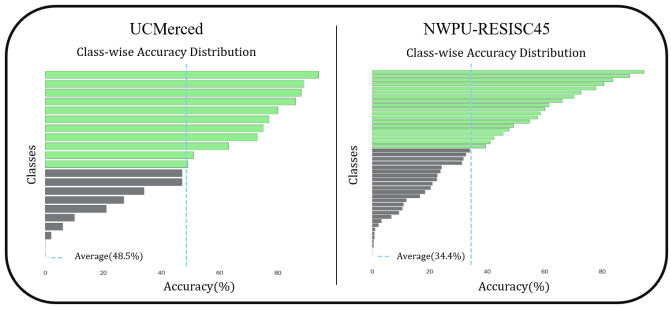


Fig. 2. Performance of CLIP on remote sensing datasets. Limited by the domain gap between remote sensing scenarios and ImageNet, the model exhibits a long-tailed distribution, and CLIP achieves suboptimal performance in specific scenarios.

To tackle the above challenges, we propose PCDA, a novel SF-UniDA framework tailored for remote sensing scene classification. PCDA is designed to transfer discriminative knowledge from a pretrained source model to an unlabeled target domain by introducing two key modules.

1) The pseudo-prototype generation module leverages a pretrained source classifier to infer class-wise confidence scores over the unlabeled target domain. For each class, the *top-K* high-confidence target samples are selected as the positive set, and their feature representations are aggregated to compute a compact class prototype in the embedding space. This approach bypasses computationally intensive source data generation [8], [9] and directly distills source domain knowledge into discriminative prototypes, thereby enhancing the practicality and efficiency of source-free adaptation.
2) The pseudo-label refinement module addresses both domain and category shifts in SF-UniDA. It introduces a discrete classifier composed of multiple independent binary classifiers, each predicting the presence of a specific source class. A target sample is assigned a binary vector pseudo-label, with a value of "1" indicating class membership. If all binary classifiers output "0," the sample is marked as unknown. To reduce domain discrepancy, we use the source classifier as a semantic reference and minimize the distance between its soft predictions and the discrete pseudo-labels, aligning target features with the source representation.

To address the class shift problem, PCDA introduces a multirecognition mechanism and adopts a multibinary classifier strategy tailored independently for each category, which discriminates target samples against all known classes individually to construct discrete pseudo-labels. Meanwhile, the source pretrained classifier is utilized to obtain soft pseudo-labels from the source domain perspective, with a global threshold for systematic discrimination. By imposing consistency constraints on the dual pseudo-labels, we not only narrow the distribution gap between the source and target domains but also solve the defect that a unified threshold fails to adapt to the heterogeneous cross-domain distributions of different categories. This provides more accurate References for unknown-class recognition and breaks through the limitation that global thresholds only rely on the statistical characteristics of the source domain, and this strategy has

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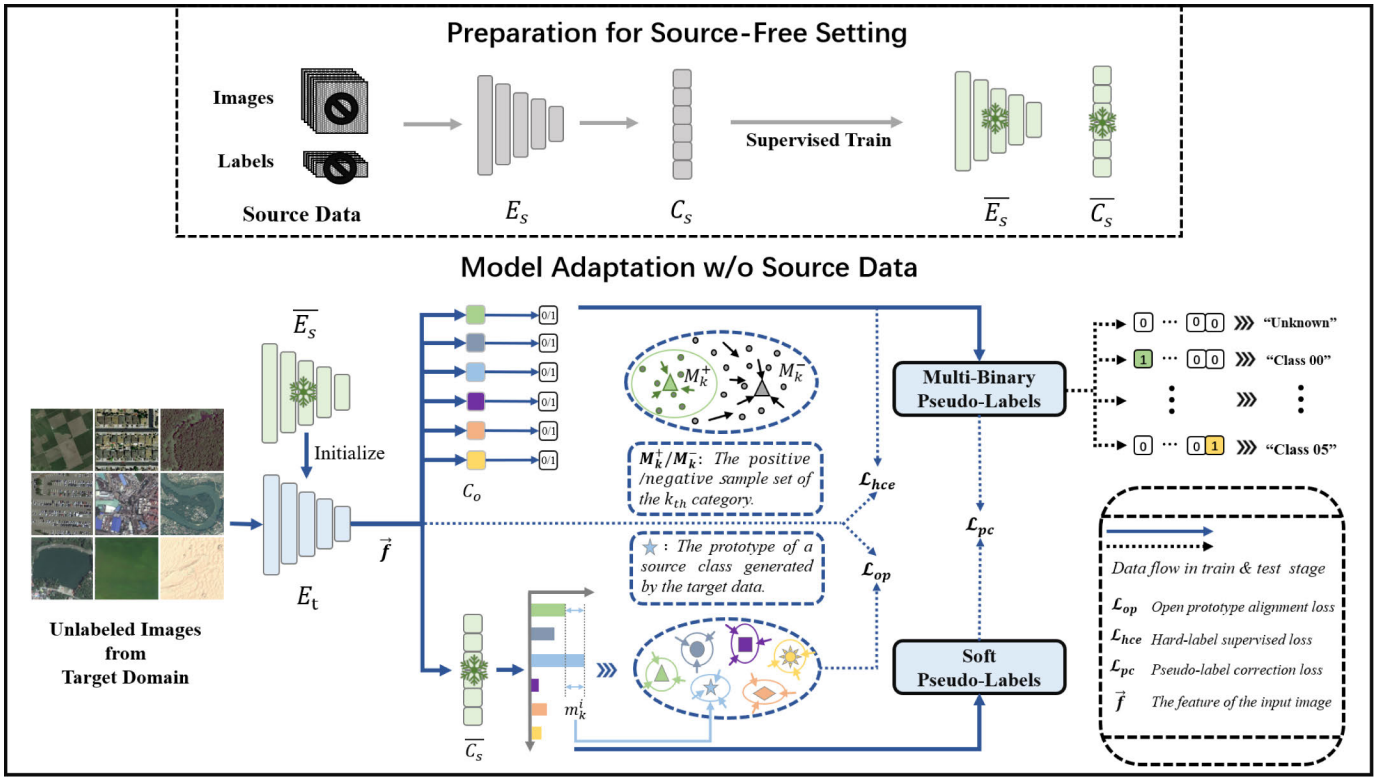


Fig. 3. Overview of the proposed network PCDA. Our framework introduces a parallel structure consisting of transferable prototype generation based on target domain data and a dual pseudo-label joint optimization module, which combines soft predictions from the source classifier and discrete binary decisions from target-specific classifiers to achieve mutually constrained learning, improving pseudo-label reliability and domain generalization. This approach addresses the challenges of SF-UniDA specifically in remote sensing scene classification tasks.

shown superior robustness and recognition accuracy in general domain adaptation (DA) scenarios [13].

To validate the effectiveness of PCDA, we utilized four remote sensing scene datasets: *Remote Sensing Scene Classification 7-Class* (RSSCN7) [14], *UC-Merced Land Use Dataset* (UCM) [15], *Aerial Image Dataset* (AID) [16], and NWPU-RESISC45 (NWPU) [17], and designed four experimental combinations, along with ablation studies.

In a nutshell, our contributions are given as follows.

1) We design a class-strategy-adaptive pseudo-prototype generation module that preserves source domain knowledge while avoiding the substantial training costs associated with explicit data generation.
2) We propose an implicit pseudo-label refinement module that introduces mutual constraints between open-domain labels and source classifier labels to mitigate interference from unknown classes and improve pseudo-label quality.
3) We propose PCDA, a novel SF-UniDA framework for remote sensing scene classification, which achieves superior performance and generalization across diverse benchmark datasets.

II. Related Work

A. DA With Source Data

Remote sensing scene classification faces two persistent challenges: the scarcity of labeled training data and distribution shifts across regions, sensors, or acquisition conditions [1], [2], [18]. To alleviate annotation burdens, DA methods have been widely adopted to transfer knowledge from labeled source domains to unlabeled target domains. Traditional approaches operate under *closed-set or partial DA* (CSDA/PDA), where the label spaces are either identical or nested. Representative methods include adversarial frameworks such as DANN [19] and GAPDA [20], which encourage domain-invariant feature extraction by aligning distributions through domain confusion. PMTrans [21] uses attention mechanisms to align multisource distributions in remote sensing.

However, these assumptions often fail in realistic remote sensing scenarios, where label sets between domains may be partially overlapping or disjoint. To address this, *open-set and universal DA* (OSDA/UniDA) methods have been proposed to distinguish shared classes and reject unknown samples. For example, OSDA-ETD [22] minimizes intraclass variance while maximizing interclass separation using transitivity-based constraints. Entropy-based rejection schemes like CMU [23] discard high-uncertainty samples as unknowns, while C^3DA [24] introduces multiple classifiers with weighted outputs to improve unknown detection. Although effective in some cases, these methods are sensitive to entropy thresholds and prone to error in imbalanced or long-tailed distributions (i.e., conditions frequently found in remote sensing scene classification tasks). *Universal DA* (UniDA) further generalizes the above by assuming no prior knowledge of label set relations. In the remote sensing category, HyUniDA [25] tackles hyperspectral UniDA via feature clustering and semantic pairing to identify shared classes.

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B. DA Without Source Data

DA methods require access to source data during adaptation, which is often impractical in remote sensing due to privacy concerns, data sensitivity, or storage constraints. This has spurred growing interest in source-free DA (SF-DA), which is achieved solely via a pretrained source model. In general vision tasks, SHOT [26] and NRC [27] employ pseudo-labeling, feature alignment, and self-training for source-free adaptation, but typically depend on batch normalization statistics or adversarial fine-tuning—strategies that face inherent limitations in remote sensing. Recent works have adapted SF-DA to remote sensing scene classification. For example, Mohammadi et al. [28] proposed an information-theoretic loss for cross-regional/cross-temporal crop classification, optimizing mutual information between predictions and target features. GLC [29] achieves SF-UniDA under dual domain and category shifts via global–local clustering and adaptive one-versus-all pseudo-labeling. More recently, LEAD [30] and GLC++ [31] enhance the robustness of semantic structure via a clustering-contrastive synergy, showing complementary advantages to our prototype+binary design in generalization and contrast. However, such high-capacity models tend to overfit under large domain shifts or ambiguous class boundaries, and entropy-based heuristics/complex clustering methods struggle to scale under long-tailed distributions and semantic uncertainty. Xu et al. [8] first introduced a two-stage SF-UniDA framework, SGD-MA, for remote sensing image classification, which synthesizes class-separable avatar data from source classifiers and aligns target features via transferable weights.

C. DA With Pseudo Labels

Within the realm of DA for RGB remote sensing image classification, RegDA [32] introduces a three-step pipeline (region division, label fusion, voting optimization), which effectively improves the quality and quantity of pseudo-labels. LFMDA [33] innovatively adopts a prototypical contrastive knowledge distillation loss to distill knowledge from generalized *vision foundation models* (VFMs), empowering the student model to acquire domain-invariant and class-discriminative features. It also integrates the segment anything model (SAM) and the LRH strategy for high-quality pseudo-label generation. For remote sensing image segmentation, Liu et al. [10] proposed a multilevel pseudo-label correction strategy, refining pseudo-labels at both category and pixel levels to enhance reliability. SAMST [34] deeply fuses SAM with a Transformer architecture and presents a progressive three-stage pseudo-label refinement pipeline. Leveraging SAM's zero-shot generalization and boundary detection, it boosts the consistency of pseudo-label spatial information and the accuracy of edge calibration.

For synthetic aperture radar (SAR) image segmentation, pseudo-labels from paired optical annotations often contain significant noise. Liu et al. [35] designed an image resolution alignment enhancement technique and introduced symmetric cross-entropy loss, effectively mitigating noise impact to improve pseudo-label quality.

Unlike prior approaches that rely on access to source data, synthetic data generation, or adversarial training, our method targets the underexplored yet practical setting of SF-UniDA for remote sensing scene classification. We address both source data inaccessibility and label space mismatch by designing a framework that avoids explicit generation and entropy-based unknown class rejection. Specifically, our method incorporates prototype-driven feature anchoring and discrete binary classifiers to robustly handle open-domain decision boundaries, achieving efficient adaptation in semantically diverse and label-ambiguous remote sensing environments.

III. Methodology

This section elaborates on the technical details of the proposed PCDA framework (as illustrated in Fig. 3) for SF-UniDA in remote sensing scene classification. PCDA is designed to address the dual challenges of domain shift and category shift while operating under strict source data privacy constraints. The core idea is to exploit the knowledge embedded in a pretrained source classifier and adapt it to an unlabeled target domain via transferable prototypes and refined pseudo-labeling strategies.

A. Problem Setting

In the SF-UniDA setting for remote sensing scene classification, we aim to adapt a source model Φ_s to an unlabeled target domain $D_t = \{x_t^i\}_{i=1}^{N_t}$, without access to source data. Here, the source and target domains differ in both data distribution ($I_s \neq I_t$) and label space ($L_s \neq L_t$). We define the label space as consisting of three parts: shared classes present in both domains, source-private classes seen only during pretraining, and target-private classes. The objective is to obtain an adapted model Φ_t that effectively recognizes shared classes while rejecting samples from unknown categories during adaptation.

B. Transferable Prototype Generation Based on Target Data

Existing prototype-based DA methods often rely on labeled source data or pseudo-labeled target samples to construct class-wise representations. However, these approaches are less suitable for SF-UniDA due to two key limitations: the inaccessibility of source samples and the potential label space mismatch between source and target domains.

To address this, we propose a target domain-driven class-strategy-adaptive prototype generation method that leverages the pretrained source classifier $\overline{C_s}$ to extract semantic priors from target predictions. Rather than relying on source data generation, we adopt a confidence-aware selection strategy that avoids the significant computational overhead incurred by generating source domain data and achieves the transfer of source domain knowledge merely by performing adaptive aggregation on unlabeled target domain data. In response to differences in knowledge transfer effectiveness across different categories, we leverage pseudo-margins [36] to implement differentiated aggregation strategies for each category, thereby enabling more refined knowledge transfer.

Formally, we define the source-side prototype matrix as

$$P_s = \left[\vec{P}_{s1}, \vec{P}_{s2}, \ldots, \vec{P}_{sk}, \quad \ldots, \vec{P}_{sn^s}\right]^\top \tag{1}$$

where n^s is the number of source classes. Denote the count of high-confidence samples under the k_{th} class as τ^k, which

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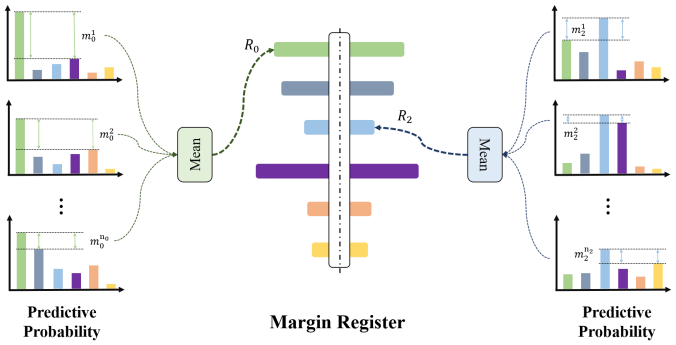


Fig. 4. Pseudo margin register: Constraining positive/negative sample sets of source classes using pseudo-margins of target samples.

is determined by the pseudo-margin register; the prediction confidence is quantified as

$$\tau^k = \frac{N_t}{h}\left(\gamma + (1-\gamma) \times \left(\frac{R_k}{2}\right)^{0.5}\right) \tag{2}$$

where h denotes the preliminary estimate of the number of classes in the target domain. To obtain h, we apply K-means clustering and evaluate the clustering quality using the Silhouette Coefficient. The Silhouette Coefficient quantifies two critical aspects of clustering performance: intracluster compactness and intercluster separation. We select the number of clusters K that yields the optimal Silhouette Coefficient (the value closest to 1) as the final value of h. R_k denotes the margin register value for the k_{th} category, which stores the average margin m_k computed overall target samples predicted as class k. To balance the influence of R_k on τ^k, the bias γ is set to 0.1 in our experiment. As shown in Fig. 4, each individual pseudo-margin m_k^i reflects the confidence gap between the top-1 and top-2 predictions for the i_{th} sample, and m_k is obtained by averaging m_k^i across the batch.

This margin-aware sample selection dynamically adjusts the size of the positive sample set M_k^+: classes with consistently low margins (typically the private categories) are downweighted to suppress noisy samples, while dominant shared classes retain larger sets. The prototype for class k is computed by aggregating the feature embeddings of the selected samples

$$\vec{P}_{sk} = \frac{1}{\tau^k} \sum_{x^t \in M_k^+} E_t\left(x^t\right) \tag{3}$$

where E_t denotes the target feature extractor.

To ensure that feature-prototype alignment reflects confidence reliability, we introduce a confidence-weighted similarity mechanism. Let $F \in \mathbb{R}^{B\times d}$ denote the features of a batch of target images and $P_s \in \mathbb{R}^{n_s\times d}$ denote the prototype matrix. The similarity matrix $Q = FP_s^\top$ captures the dot-product similarity between features and prototypes. For each sample–prototype pair, we define the normalized confidence weight

$$A\left(\vec{P}_{sk}, \vec{f}_j\right) = \frac{\exp\left(Q\left(\vec{P}_{sk}, \vec{f}_j\right)\right)}{\sum_{k'} \exp\left(Q\left(\vec{P}_{sk'}, \vec{f}_j\right)\right)} \tag{4}$$

where the k' denotes all source categories other than the k_{th} class. The corresponding distance between prototype $\vec{P}_{sk}$ and target feature $\vec{f}_j$ is measured using cosine distance

$$D\left(\vec{P_{sk}}, \vec{f_j}\right) = 1 - \frac{\vec{f_j}^T \vec{P_{sk}}}{\|\vec{f_j}\| \cdot \|\vec{P_{sk}}\|}. \tag{5}$$

Finally, the open set prototype-based alignment loss is defined as the confidence-weighted distance between each sample and all prototypes

$$L_{op} = \sum_{k}^{n^s} \sum_{j}^{N_t} A\left(\vec{P}_{sk}, \vec{f}_j\right) D\left(\vec{P}_{sk}, \vec{f}_j\right) \tag{6}$$

where $A(\cdot,\cdot)$ is used to calculate the confidence between the positive prototype $\vec{P}_{sk}$ and feature vector $\vec{f}_j$, while $D(\cdot,\cdot)$ denotes the distance between $\vec{P}_{sk}$ and $\vec{f}_j$.

C. Pseudo-Label Learning Based on Specific Binary Classifiers

Building upon the transferable prototype module, which leverages high-confidence target data to construct class-level representations aligned with the source label space, we further introduce a pseudo-label learning module to support open-domain recognition under label space mismatch. While prototypes provide semantic anchors for known categories, effective discrimination in SF-UniDA also requires identifying whether a target sample truly belongs to the shared label space or represents the unknown class.

To explicitly resolve the challenges of class shift and domain shift without access to source data, we propose a pseudo-label learning module built upon a group of category-specific binary classifiers. This design enables us to generate discrete semantic labels for target samples, supporting open-domain discrimination where unknown classes may appear.

We begin by dividing target samples into two conceptual groups based on their potential semantic alignment with the source domain: public-class samples, which have corresponding categories in the source domain, and private-class samples, which do not. While shared-class samples can potentially inherit semantic labels from the source classifier $\overline{C_s}$, direct predictions may be unreliable due to domain shift. To overcome this, we construct a set of binary classifiers $C_o = \{C_o^1, \ldots, C_o^{n^s}\}$, each trained to identify the presence or absence of a specific source class. These classifiers collectively serve as an open-domain labeling mechanism, outputting binary vectors where each entry indicates membership "1" or nonmembership "0" with respect to a given class.

For a specific category k, we first construct a positive sample set M_k^+ by selecting target samples with high confidence scores from $\overline{C_s}$, and compute a corresponding positive prototype $\vec{P}_{sk}$. All other target samples are treated as forming the negative set M_k^-, which may include samples from other classes or ambiguous cases of class k. To robustly summarize this set, we apply a clustering algorithm to obtain a negative prototype $\vec{N}_{sk}$, capturing the central tendency of nonclass-k examples. By introducing the *K-means* clustering algorithm to aggregate information from the negative sample set, the negative prototype $\vec{N}_{sk}$ can be obtained as follows:

$$\vec{N}_{sk} = Kmeans_{x^t \in M_k^-}\left(E_t\left(x^t\right)\right) \tag{7}$$

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where the number of negative prototypes under the kth category follows the number of clusters h from the previous step. Each binary classifier C_o^k determines class membership by computing the cosine distances between a target sample and the corresponding positive and negative prototypes. The hard pseudo-label q_k^j for target sample x_t^j is generated according to the following rule:

$$q_k^j = \begin{cases} 1, & if\ D\left(\vec{f}_j, \vec{P}_{sk}\right) \leq \min\left\{D\left(\vec{f}_j, \vec{N}_{sk}^i\right)\right\}_{i=1}^{h} \\ 0, & if\ D\left(\vec{f}_j, \vec{P}_{sk}\right) > \min\left\{D\left(\vec{f}_j, \vec{N}_{sk}^i\right)\right\}_{i=1}^{h} \end{cases} \tag{8}$$

where D(•,•) is used to compute the cosine distance between the sample feature and the prototypes. Then each target sample is assigned a binary vector q^j composed of "0" and "1." A one-hot vector indicates that the sample is confidently associated with a specific known class, while a zero vector signifies that the sample has been rejected by all source class classifiers and is thus identified as belonging to an unknown category.

Compared to direct use of source classifier $\overline{C_s}$, this binary-classifier-based approach integrates target-domain characteristics into decision boundaries, improving robustness to domain shifts and class ambiguity. Additionally, it avoids reliance on global entropy thresholds, which can be unreliable in imbalanced or fine-grained remote sensing datasets. To supervise model training, we first define a hard label cross-entropy loss $\mathcal{L}_{hce}$ using the generated pseudo-labels

$$\mathcal{L}_{hce} = -\frac{1}{N_t}\sum_{j=1}^{N_t}\sum_{k=1}^{h} q_k^j \log\left(\overline{C_s^k}\left(\vec{f}_j\right)\right). \tag{9}$$

While the discrete binary labels are effective in rejecting unknown classes, their performance on known classes can be further improved. To this end, we incorporate the soft predictions from the frozen source classifier $\overline{C_s}$ as auxiliary supervision and enforce consistency between the discrete predictions and the soft outputs via a Kullback–Leibler (KL) divergence loss

$$\mathcal{L}_{pc} = D_{KL}\left(\log\left(\overline{C_s}\left(\vec{f}_j\right)\right), \vec{q^j}\right). \tag{10}$$

During the training stage, by combining the cross-entropy loss function with the entropy constraint, the model learns more smoothly. Additionally, by assigning a lower participation weight to private class samples, we can fully utilize these samples for training, which enhances the model's understanding and generalization to target samples. Therefore, the final objective of our global semantic supervision is adjusted to

$$\mathcal{L}_{pc}^{*} = \mathcal{L}_{hce} + \lambda \cdot \mathcal{L}_{pc} \tag{11}$$

where λ, which balances the contribution of soft-label alignment and is found to be optimal at 0.3 for the task. By jointly optimizing the pseudo label correction loss $\mathcal{L}_{pc}^{*}$ and the prototype-based alignment loss $\mathcal{L}_{op}$ introduced earlier, the total objective becomes

$$\mathcal{L}_{total} = \mathcal{L}_{pc}^{*} + \mathcal{L}_{op}. \tag{12}$$

This formulation enables robust semantic learning across domains, effectively handling both domain shift and class shift under the SF-UniDA setting for remote sensing scene classification.

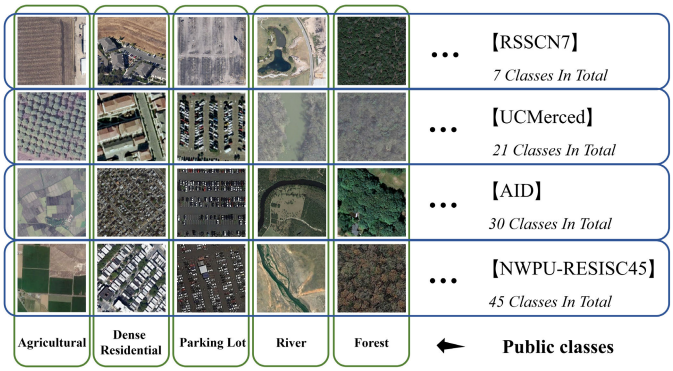


Fig. 5. Four publicly recognized remote sensing benchmarks: RSSCN7, UCM, AID, and NWPU, which were adopted to validate the proposed PCDA framework across diverse scenarios.

IV. Experiments

We provide a comprehensive description of the experimental design, including comparative experiments, ablation studies, and parameter sensitivity analysis. These experiments are intended to validate the practical feasibility and effectiveness of the proposed method, as well as to assess its performance under various conditions.

A. Experimental Setup

This section will provide a detailed description of the experimental setup, including the datasets used and implementation details. By clearly outlining the experimental conditions, we aim to ensure the reproducibility and objectivity of the experiments.

1) Datasets: To validate our method, we selected four well-known cross-domain remote sensing scene datasets (as shown in Fig. 5): RSSCN7 [14], UCM [15], AID [16], and NWPU [17]. The RSSCN7 dataset consists of 2800 images with a resolution of 400 × 400 pixels, spanning seven typical scene categories, such as forest, grassland, and agriculture. These images were captured under varying seasonal and weather conditions. The UCM dataset contains 2100 images, each with a resolution of 256 × 256 pixels, classified into 21 scene categories, including airports and agricultural areas. This dataset is widely used in remote sensing classification research. The AID dataset, sourced from Google Earth, comprises 10 000 images with a resolution of 600×600 pixels, covering 30 scene categories (including airports and ports), and is suitable for open-domain classification and object detection tasks. Finally, the NWPU dataset, one of the largest available remote sensing image datasets, includes 31 500 images with a resolution of 256×256 pixels, categorized into 45 scene classes. This dataset demonstrates substantial diversity in terms of temporal, spatial, and lighting variations, making it ideal for classification and retrieval tasks.

As shown in Table I, there are four different settings of UniDA. RSSCN7 has only 7 categories, whereas the other three datasets have 21, 30, and 45 categories, respectively. Therefore, transferring from RSSCN7 to the other three datasets, to a certain extent, reflects the difficulty of DA tasks based on the differences in the number of categories. Additionally, we conduct experiments using AID as the source

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TABLE I
FOUR UNIDA TASKS FOR REMOTE SENSING SCENE CLASSIFICATION

RSSCN7 → UCM	Total label sets	Shared label sets	Private label sets	ξ
Source domain :RSSCN7 dataset	7	5	2	0.18
Target domain :UC Merced dataset	21		16	
RSSCN7 → AID	Total label sets	Shared label sets	Private label sets	ξ
Source domain :RSSCN7 dataset	7	6	1	0.16
Target domain :AID dataset	30		24	
RSSCN7 → NWPU	Total label sets	Shared label sets	Private label sets	ξ
Source domain :RSSCN7 dataset	7	6	1	0.12
Target domain :NWPU-RESIS45 dataset	45		39	
AID → NWPU	Total label sets	Shared label sets	Private label sets	ξ
Source domain :AID dataset	30	20	10	0.27
Target domain :NWPU-RESIS45 dataset	45		25	

domain and NWPU as the target domain to evaluate the performance of our method on larger datasets. We use ξ to measure the gap between the label sets of the source and target domains, which can be calculated using the following equation:

$$\xi = \frac{n_c}{n^s + n^t} \tag{13}$$

where the notation n_c denotes the number of shared classes, while n^s represents the number of source domain classes, and n^t represents those of the target domain.

2) Implementation Details: The overall training procedure comprises two distinct phases: the source model preparation stage and the SF-UniDA stage. In the preparation stage, supervised learning is conducted on the source domain data for 30 epochs, and the corresponding model parameters are saved as $\overline{E_s}$ and $\overline{C_s}$. During the SF-UniDA stage, $\overline{E_s}$ is utilized to initialize the target domain feature extractor E_t, followed by 50 epochs of unsupervised learning on the target domain data. Throughout the training process, standard data augmentation techniques such as center cropping and horizontal flipping are applied. All input images are first resized to 256×256 pixels, and then center-cropped to 224×224 pixels.

Stochastic gradient descent (SGD) with momentum was used as the optimizer, where the momentum was set to 0.9 and weight decay was fixed at 0.001. To facilitate DA, distinct learning rates are assigned to the source and target domains: 0.001 for the source domain and 0.0001 for the target domain. The batch size was uniformly set to "128" across all of the SF-UniDA experiments. To ensure experimental reproducibility, the random seed is fixed to "1017." All methods involved in the experiments adopt ResNet-50 pretrained on the ImageNet dataset [37] as the backbone network, except for "Ours(ConvNeXt)," which utilizes a ConvNeXt-Base [38] backbone and "Ours(ViT) which utilizes a ViT-Base backbone".

B. Comparative Experiments

We present a comprehensive evaluation of the proposed PCDA framework for SF-UniDA under remote sensing scene classification. To strengthen the comparative validity of our experiments, we developed variants of the proposed method with different backbones. Specifically, Ours(ConvNeXt) adopts the pretrained ConvNeXt-Base model as the backbone, and Ours(ViT) uses the ViT-Base model as the backbone with a patch size set to 16 pixels.

1) Competitors: To comprehensively analyze the improvements of our proposed PCDA, we introduce some methods under this task. RS-CLIP is introduced to show the performance of methods based on CLIP. What's more, the *state-of-the-art* (SOTA) approach under SF-UniDA setting (SGD-MA) is also demonstrated. Similarly, we incorporate the latest SF-UniDA methods in the natural image domain (such as GLC, LEAD, and GLC++) to provide a more comprehensive comparative perspective. To establish a standardized benchmark, all comparative methods adopt ResNet-50 as the backbone, including the visual encoder of CLIP.

RS-CLIP [39] is a zero-shot remote sensing scene classification method that leverages contrastive vision-language supervision, enabling accurate classification of unseen remote sensing scene categories without task-specific labeled training data for the target categories.

GLC [29] is a method for upcycling models under domain and category shift, which tackles the dual challenges of domain distribution discrepancy and category set mismatch to reuse pretrained models effectively without relying on additional source data for adaptation.

SGD-MA [8] is the first SF-UniDA method for remote sensing image classification, comprising source data generation and model adaptation stages. It synthesizes source data from pretrained models and uses transferable weights to distinguish shared/private label sets for classifying target samples and identifying unknown categories.

LEAD [30] is a SF-UniDA method leveraging learning decomposition, which addresses domain shift and category gap challenges in the source-free setting to better align source and target domain feature distributions without source data.

GLC++ [31] is a SF-UniDA method that adopts global–local clustering and integrates contrastive affinity learning, enabling accurate classification of known data, segregation of unknown data, and improved novel category clustering accuracy.

2) Result on RSSCN7 → UCM: In this experiment, we select the RSSCN7 dataset as the source domain and the

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TABLE II
CLASSIFICATION ACCURACY OF UNIDA WITHOUT SOURCE DATA ON RSSCN7 → UCM(%)

RSSCN7 → UCM	Agricultural	Dense Residential	Forest	Parking Lot	River	Unknown	Average
Clip	34.00	2.00	95.00	97.00	**77.00**		61.00
RS-Clip	66.00	27.00	95.00	75.00	70.00		66.60
GLC	42.11	46.34	100.0	100.0	74.39	15.16	72.60
SGD-MA	92.00	63.00	64.00	93.00	76.00	17.13	77.60
LEAD	**96.05**	48.78	100.0	90.67	65.85	19.52	80.30
GLC++	86.91	59.21	100.0	65.12	70.37	17.90	76.32
Only Source	87.50	22.10	99.00	91.90	44.40	15.60	69.20
Ours	95.80	**93.30**	**100.0**	**100.0**	61.10	**25.20**	**88.10**
Ours(ConvNeXt)	100.0	94.74	100.0	100.0	62.50	55.14	91.40
Ours(ViT)	99.00	80.00	90.00	100.0	96.00	56.94	93.20

TABLE III
CLASSIFICATION ACCURACY OF UNIDA WITHOUT SOURCE DATA ON RSSCN7 → AID(%)

RSSCN7 → AID	Agricultural	Dense Residential	Forest	Parking Lot	River	Industrial Area	Unknown	Average
Clip	62.70	30.70	45.20	94.60	27.60	**89.70**		58.42
RS-Clip	85.14	60.73	92.80	98.46	42.20	58.72		73.01
GLC	86.76	62.20	94.00	45.13	26.34	52.57	10.33	59.50
SGD-MA	**97.80**	78.80	69.20	96.90	37.30	62.10	17.90	73.60
LEAD	60.66	89.74	100.0	98.02	**79.76**	15.56	04.19	72.30
GLC++	93.24	60.24	100.0	13.59	41.46	67.69	11.44	62.71
Only Source	71.20	82.40	96.40	57.60	53.90	71.40	22.34	72.20
Ours	97.50	**97.60**	**100.0**	**98.50**	35.80	67.10	**27.12**	**82.70**
Ours(ConvNeXt)	96.34	82.87	99.50	98.11	64.51	58.22	16.93	83.30
Ours(ViT)	88.11	83.17	99.20	95.90	62.93	76.15	30.81	84.20

UCM dataset as the target domain. All experimental results are presented in Table II: "Unknown" is defined as the accuracy of the model in identifying unknown classes, while "Average" refers to the average accuracy of the public classes. RS-CLIP [39] optimized text prompt design and cross-modal alignment strategies for remote sensing-specific categories (e.g., "Agricultural," "Forest") to overcome the limitations of the original CLIP model in remote sensing semantic understanding. This achieves a 5.6% improvement compared to CLIP's baseline (61.00%). SGD-MA, the first SF-UniDA method in the remote sensing scene image domain, achieves an average accuracy of up to 77.6%, along with a 17.13% unknown-class rejection rate.

In contrast, our method achieves an average accuracy of 88.10% (a +10.5% gain over SGD-MA) and a 25.20% unknown-class rejection rate (an +8.07% improvement). In terms of overall performance: existing methods from the natural image domain GLC, LEAD, GLC++ yield average accuracies of 72.60%, 80.30%, and 76.32%, respectively. PCDA outperforms these methods and achieves 100% accuracy in both "Forest" and "Parking Lot" categories. This superiority stems from our method's structural semantic anchors (prototypes) and category-aware open set binary classifiers, which jointly model shared-class semantic consistency and unknown-class exclusivity, thus enhancing discriminative ability and robustness under domain/category shifts.

3) Result on RSSCN7 → AID: In this experiment, we select the RSSCN7 dataset as the source domain and the AID dataset as the target domain. All experimental results are presented in Table III. The challenge intensifies notably due to a low category overlap ratio ($\xi = 0.16$) and enhanced semantic divergence between the source and target domains.

In the field of remote sensing scene image classification, the SOTA method under the SF-UniDA paradigm (SGD-MA) achieves a classification accuracy of 73.6%. This result not only outperforms the CLIP-based zero-shot method RS-CLIP (73.01%) but also significantly surpasses methods from the natural image domain such as GLC, LEAD, and GLC++. Compared with SGD-MA, "Ours" further improves the classification accuracy by 9.10%, and even achieves 100% classification accuracy in the "Forest" category. In addition, we develop two extended variants based on our proposed PCDA method, named Ours (ConvNeXt) and Ours (ViT), which use ConvNeXt-Base and ViT-Base as the backbone network, respectively. Compared with the 88.1% average accuracy yielded by the baseline model, our two variants achieve absolute improvements in average accuracy of 0.6% points and 1.5% percentage points, respectively.

4) Result on RSSCN7 → NWPU: In this experiment, we select the RSSCN7 dataset as the source domain and the NWPU dataset as the target domain. The classification accuracy of PCDA and comparative approaches on the SF-UniDA task is reported in Table IV.

CLIP (with ResNet-50 as the image encoder) achieves only 54.26% mean accuracy on NWPU. RS-CLIP, which designs domain-specific prompts tailored for remote sensing scene

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TABLE IV
CLASSIFICATION ACCURACY OF UNIDA WITHOUT SOURCE DATA ON RSSCN7 → NWPU(%)

RSSCN7 → NWPU	Agricultural	Dense Residential	Forest	Parking Lot	River	Industrial Area	Unknown	Average
Clip	60.30	38.40	41.00	**70.30**	01.00	60.30		54.26
RS-Clip	69.57	59.00	92.14	86.43	39.00	43.86		65.00
GLC	84.29	78.14	99.14	15.71	30.07	82.71	09.93	65.01
SGD-MA	**91.86**	**88.00**	86.57	28.00	**84.86**	65.86	**25.20**	**74.19**
LEAD	82.33	16.51	**99.82**	14.87	09.41	87.88	13.68	52.80
GLC++	81.99	87.86	99.43	38.43	36.29	78.00	10.19	70.33
Only Source	58.21	63.39	94.96	48.15	54.01	**78.23**	15.67	65.90
Ours	70.15	86.61	97.84	42.96	63.87	68.71	18.39	71.70
Ours(ConvNeXt)	89.29	45.52	100.0	63.24	63.76	76.71	26.97	73.10
Ours(ViT)	83.71	80.57	95.71	81.43	52.93	57.29	30.06	75.30

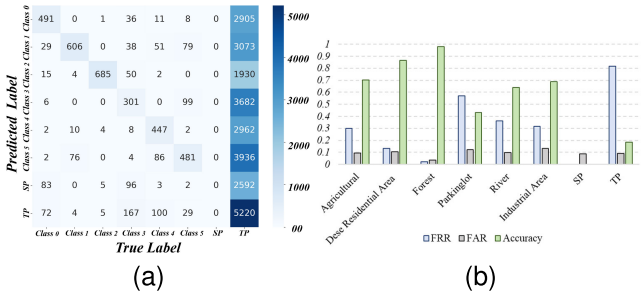


Fig. 6. RSSCN7 → NWPU failure cause analysis. (a) Confusion matrix. (b) FRR, FAR, and overall accuracy; SP: Source private class; TP: Target private class.

images, effectively improves the classification accuracy of CLIP on remote sensing scene image classification tasks. However, compared with such zero-shot methods, SF-UniDA methods yield more significant task-specific performance gains under the setting of source domain inaccessibility. In particular, "Ours" outperforms existing methods (GLC, LEAD, and GLC++) in the natural image domain in both the unknown-class sample rejection success rate and the mean accuracy of public classes. "Ours" follows the standardized benchmark setup with ResNet-50 as the backbone, fully aligned with all comparative methods. On this task, our method shows consistent performance gains across variants: Ours, Ours (ConvNeXt), and Ours (ViT) achieve average classification accuracies of 71.70%, 73.10%, and 75.30%, respectively, among which Ours (ViT) outperforms the well-established SGD-MA (74.19%) by an absolute 1.11% point.

Although PCDA achieves a slightly lower average accuracy than the comparison method SGD-MA on this task, we argue that this does not indicate a methodological weakness. Instead, the performance gap reflects differences in adaptation strategies under extremely low-overlap scenarios.

As shown in Fig. 6(a), the misclassification of samples from public categories is mainly disturbed by the private classes of the target domain, and the number of falsely accepted samples between public categories is not high, indicating that the model's classification ability on public category samples is as expected. According to the prediction results of SP, the number of predicted samples of the private classes in the source domain is very small, suggesting that they have been effectively suppressed. Meanwhile, as shown in Fig. 6(b), the false rejection rate (FRR) of TP exceeds 0.8, meaning numerous unknown samples are misclassified as known classes. By contrast, the average false acceptance rate (FAR) of shared classes remains below 0.1, proving strong discriminability between shared categories. Including TP samples pushes the overall FAR above 0.8, verifying that performance degradation stems mainly from TP interference.

What's more, PCDA relies on a small number of shared classes to construct transferable prototypes, which can be more vulnerable to semantic contamination when shared samples are scarce. In contrast, SGD-MA benefits from the class separability of synthesized pseudo-source data, providing a stronger discriminative signal during early adaptation. Nevertheless, PCDA avoids synthetic data generation while still demonstrating strong capabilities in feature alignment and semantic structuring. Moreover, it achieves more stable and generalizable performance across other transfer tasks. Therefore, we regard PCDA as methodologically comparable to *SGD-MA* and interpret the observed performance gap as task-specific rather than indicative of algorithmic inferiority.

5) Result on AID → NWPU: In this experiment, the AID dataset is selected as the source domain and the NWPU dataset as the target domain. Both PCDA and comparative approaches on the SF-UniDA task are reported in Table V.

Compared with existing CLIP-based zero-shot methods and SF-UniDA methods, proposed PCDA ("Ours") still achieves a significant performance breakthrough. Specifically, "Ours" delivers a 5.5% accuracy improvement over the "Only Source" pretrained model (which attains 70.10% accuracy). As shown in Fig. 7(d), different categories (represented by different colors) in the "Only Source" are clustered together and difficult to distinguish. After applying our PCDA framework, the results Fig. 7(h) show that different categories are clearly dispersed, while the private class (gray data points) converges, closely aligning with the optimization objective of SF-UniDA. Among SF-UniDA methods in the natural image domain, GLC++ reaches a mean accuracy of 74.74%, yet our PCDA outperforms the SOTA method by 0.86%. Additionally, we introduce two PCDA variants, "Ours (ConvNeXt)" (with ConvNeXt-Base backbone) and "Ours (ViT)" (with ViT backbone), which achieve accuracies of 82.20% and 78.80%, respectively.

6) Analysis of Label Overlap and Pseudo-Label Reliability: To further analyze the influence of category overlap on model performance, we conduct a controlled study on the AID →

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TABLE V

CLASSIFICATION ACCURACY OF UNIDA WITHOUT SOURCE DATA ON AID → NWPU(%). AG:AGRICULTURAL, AI:AIRPORT, BF:BASEBALL FIELD, BE:BEACH, BR:BRIDGE, CH:CHURCH, CA:COMMERCIAL AREA, DR:DENSE RESIDENTIAL, DE:DESERT, FO:FOREST, IA:INDUSTRIAL AREA, ME:MEADOW, MR:MIDDLE RESIDENTIAL, MO:MOUNTAIN, PL:PARKING LOT, RS:RAILWAY STATION, RI:RIVER, SR:SPARSE RESIDENTIAL, ST:STADIUM, ST:STORAGE TANKS

AID → NWPU	Ag	Ai	BF	Be	Br	Ch	CA	DR	De	Fo	IA	Me	MR	Mo	PL	RS	Ri	SR	St	ST	Unknown	Avg
Clip	60.30	32.60	47.70	31.70	22.40	00.40	54.90	39.40	45.60	41.00	60.00	00.00	61.60	02.10	70.30	06.60	01.00	58.60	49.10	09.30		34.48
RS-Clip	69.57	51.00	58.43	51.71	55.29	32.00	12.43	59.00	88.29	**92.14**	43.86	21.43	37.43	74.00	86.43	44.57	39.00	35.43	75.57	54.43		54.10
GLC	**94.59**	**63.38**	**96.40**	**95.86**	86.36	53.52	37.21	78.06	90.71	78.26	57.42	**95.69**	44.76	32.48	91.20	16.92	**85.00**	83.57	91.30	84.56	23.20	72.90
SGD−MA	80.57	44.86	81.00	79.57	94.00	52.86	41.57	29.57	**91.71**	79.43	82.29	88.57	20.57	12.00	83.71	73.00	77.00	90.43	62.71	78.71	20.22	64.97
LEAD	90.40	01.17	89.48	89.19	66.53	03.72	52.01	**84.34**	85.54	90.93	53.03	83.91	73.97	**95.95**	87.48	79.26	38.39	**90.89**	**92.17**	91.84	16.84	72.00
GLC++	88.29	43.14	75.71	81.14	82.86	45.14	50.29	82.71	87.14	91.57	72.29	80.43	58.86	73.29	90.29	**81.00**	70.86	55.57	89.43	94.86	28.13	74.74
Only Source	89.86	28.29	56.86	68.28	84.86	**74.57**	49.57	74.14	75.71	84.14	72.86	69.86	76.86	79.57	84.00	69.86	57.14	65.00	45.71	94.14	20.12	70.10
Ours	90.85	38.57	84.17	89.17	**94.52**	71.76	**66.91**	66.43	74.58	62.77	**86.81**	84.29	**78.32**	41.30	**91.94**	57.79	77.44	76.38	82.98	**95.83**	**33.80**	**75.60**
Ours(ConvNeXt)	92.75	55.02	79.32	89.55	94.02	81.01	54.29	81.47	83.21	94.84	75.41	77.05	85.46	89.87	89.57	88.83	71.96	79.64	85.77	95.92	11.42	82.20
Ours(ViT)	92.43	56.57	71.29	91.57	82.86	75.43	60.86	83.00	81.00	82.43	80.29	77.57	48.29	91.57	83.71	80.00	82.86	78.00	80.29	94.57	18.43	78.70

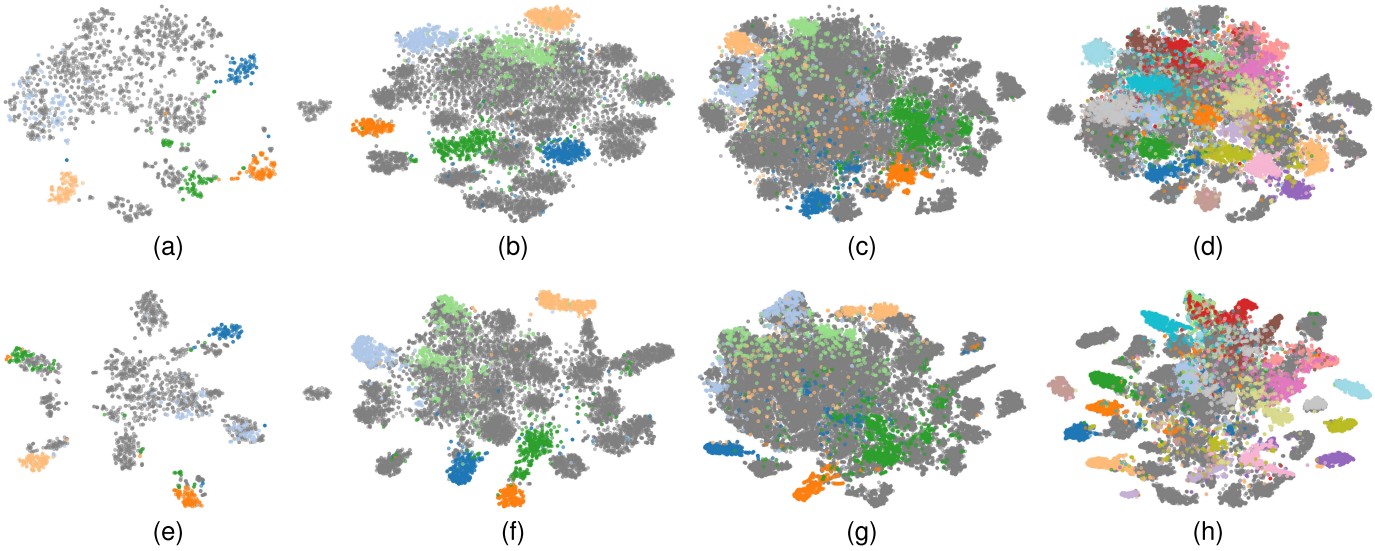


Fig. 7. Visualization results of sample feature distributions before and after applying the proposed method. Subfigures (a)–(d) Display the sample feature distributions of the four dataset combinations (R2U, R2A, R2N, and A2N) prior to the application of the proposed method. Subfigures (e)–(h) Present the corresponding sample feature distributions after the implementation of PCDA. As shown in these subfigures, the clustering effect of sample clusters from different categories (Gray dots represent unknown classes) exhibits a noticeable improvement. (a) R2U-source only. (b) R2A-source only. (c) R2N-source only. (d) A2N-source only. (e) R2U-trained with PCDA. (f) R2A-trained with PCDA. (g) R2N-trained with PCDA. (h) A2N-trained with PCDA.

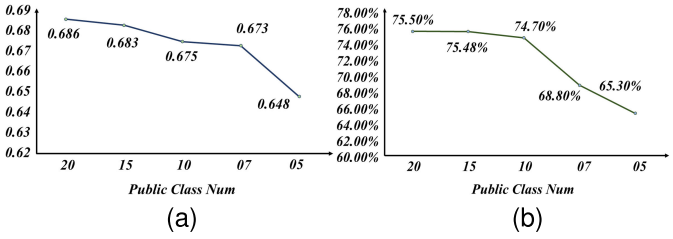


Fig. 8. Analysis of the impact of shared category overlap on DA performance. (a) H-score. (b) Accuracy under different levels of shared classes in AID → NWPU.

NWPU transfer task. Specifically, we design multiple experimental settings by systematically re-partitioning the label spaces of the two datasets, gradually decreasing the number of shared categories. This setup enables us to simulate SF-UniDA scenarios with varying overlap levels and observe the corresponding changes in model performance.

As illustrated in Fig. 8, both the average accuracy and H-score exhibit a consistent upward trend as the number of shared classes increases. This observation supports our hypothesis that semantic diversity in the source label space plays a vital role in enhancing the model's generalization to the target domain. The findings further emphasize the necessity of constructing diverse and semantically rich source label sets to achieve effective adaptation in large-scale, heterogeneous remote sensing environments.

C. Ablation Study

The adaptive pseudo-prototype generation module proposed in PCDA constrains the generation of prototypes through L_{op}. Meanwhile, the pseudo-label correction module constrains the two pseudo-labels via L_{pc} for joint optimization. As shown in Table VI, ablation experiments are conducted on four datasets to quantify the contributions of these two modules. Only Source denotes the source-only model (without access to target domain data) and reports the average accuracy of public classes and the recognition accuracy of unknown classes across these four datasets. "Ours" represents the performance of our proposed PCDA method. Notably, in the ablation experiment on RSSCN7 → UCM, "Ours" achieves an average accuracy of 88.10%, which constitutes a substantial improvement of 18.9% compared with "Only Source" (69.20%), while the recognition accuracy of unknown classes is also enhanced by 9.6%. Similarly, on the

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TABLE VI
RESULT OF ABLATION EXPERIMENT CONDUCTED ON THE FOUR COUPLES OF DATASET

Ablation Result	RSSCN7 → UCM		RSSCN7 → AID		RSSCN7 → NWPU		AID → NWPU	
	Average	unknown	Average	unknown	Average	unknown	Average	unknown
Ours $w/o\ L_{op}\ \&\ L_{pc}$	77.20	20.40	70.90	23.42	65.90	10.47	72.10	25.65
Ours $w/o\ L_{pc}$	80.30	23.60	78.30	24.44	67.80	14.93	74.60	29.34
Ours	**88.10**	**25.20**	**82.70**	**27.12**	**71.70**	**18.39**	**75.60**	**33.80**

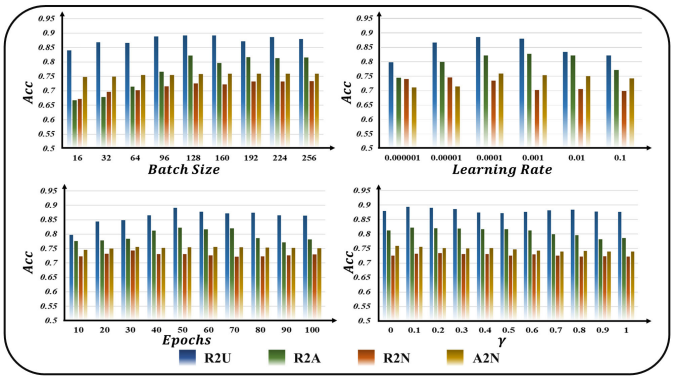


Fig. 9. Hyperparameter sensitivity analysis. Performance variation of the model in response to changes in batch size, learning rate, number of training epochs, and the coefficient γ.

RSSCN7 → AID dataset, removing the L_{pc} significantly impairs the model's ability to constrain open-set pseudo-labels, leading to a 4.4% drop in average accuracy. Additionally, releasing the constraint of L_{op} on pseudo-prototypes results in an 11.80% decrease in average accuracy relative to "Ours" (82.70%), and the model's capability to recognize unknown-class samples is also notably affected, causing a 3.70% reduction in the recognition accuracy of unknown classes. Furthermore, the ablation experimental results on RSSCN7 → NWPU and AID → NWPU exhibit the aforementioned trend. These data collectively demonstrate that the two proposed modules make significant contributions to the task.

To more intuitively demonstrate our core contributions, we extracted features from test images using both the DA-untreated model (referred to as the "Source Only" model) and the PCDA-optimized model (referred to as "PCDA") on four dataset pairs, followed by dimensionality-reduced visualization, as shown in Fig. 7. Particularly in Fig. 7(d) and (h), each shared class is effectively distinguished and separated from the private class (in gray). This exhibits a significant improvement compared to the original model, where samples of different classes are mixed, indirectly demonstrating the effectiveness of the pseudo-label correction strategy.

D. Parametric Sensitivity Analysis

We present the tuning and visualization results of several key parameters mentioned in our experiments, with all parameter studies conducted on all of the four dataset pairs mentioned in the experiment section. Fig. 9 displays the variation trends of the average accuracy with respect to training epochs, batch size, learning rate, and the bias variable γ from (2), respectively.

It can be observed that as the batch size increases from 16 to 128, the classification accuracy consistently exhibits an overall slight upward trend across all four datasets. Increasing the batch size reduces the variance of pseudo-label generation and enhances the stability of the pseudo-label correction module in PCDA. When the batch size exceeds 128, the model performance enters a saturation regime, and a slight degradation is observed on different datasets at batch sizes of 160 and 192, respectively. This is attributed to the high dimensionality and strong scene specificity of remote sensing images: using an excessively large batch size causes the model to fail to adequately capture fine-grained sample details. Moreover, the absence of source data for calibration in the source-free scenario increases the risk of overfitting.

The learning rate affects training by regulating the parameter update step size of gradient descent; an excessively high learning rate can cause the model parameters to update too drastically, while PCDA relies on the implicit distillation of source domain information from the source domain model. We adopt linear decay to regulate the training process; thus, the number of epochs exerts a similar yet slightly different impact compared to the initial learning rate.

It can be clearly observed from the learning rate ablation results in Fig. 9 that the final performance of the model is strongly governed by the initial learning rate, as either an excessively high or an excessively low learning rate will induce a drop in classification accuracy. Across the four datasets, the peak classification accuracy is consistently achieved at an initial learning rate of approximately $1e^{-4}$, with minor discrepancies arising from variations in the domain gap across datasets.

To determine the optimal parameter regulating the influence of the pseudo-margin on positive sample set selection, a bias term γ is incorporated into (2). As γ increases, the efficacy of the pseudo-margin in facilitating the aggregation of the positive sample set is attenuated. Conversely, if γ is excessively small, the pseudo-margin formulated based on confidence metrics becomes susceptible to noisy perturbations, thereby resulting in significant degradation of model performance. To conclude, PCDA shows negligible performance variations in response to parameter changes, which confirms that the model is not sensitive to parameter fluctuations.

V. CONCLUSION

In this article, we propose a lightweight SF-UniDA framework named "PCDA" to address the challenges of category shift and domain discrepancy in remote sensing scene classification. Instead of relying on source data generation or large-scale pretrained models, PCDA transfers knowl-

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edge through class prototypes and constructs category-specific binary classifiers to derive discrete pseudo-labels under open-set conditions. These labels are further refined by a pseudo-label correction mechanism to enhance adaptation performance across diverse target domains. Experimental results on four cross-domain settings demonstrate that PCDA achieves competitive or superior performance compared to existing methods, including those requiring significant computational overhead. Notably, PCDA performs robustly even without access to source data, validating its practicality in privacy-sensitive and resource-constrained scenarios.

While PCDA achieves consistent improvements across a range of DA tasks, certain limitations remain under highly imbalanced or mismatched label space conditions. For example, in the RSSCN7 → NWPU, the significant semantic gap and fine-grained category structure of the target domain hinder the reliability of pseudo-label refinement, which in turn weakens feature alignment. These results reflect the intrinsic challenges of SF-UniDA in complex remote sensing scenarios. Looking ahead, our framework provides a promising foundation for scalable and efficient remote sensing adaptation. In summary, PCDA provides a lightweight, complementary solution for remote sensing scene classification tasks and offers a more flexible technical option for SF-UniDA research. Future work could explore incorporating knowledge from large-scale pretrained models to enhance adaptability while preserving the lightweight and generalizable nature of the current design, ultimately advancing toward more practical and comprehensive solutions for remote sensing classification.

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